

AI-Based Garbage Detection System

Using Convolutional Neural Network (CNN) for Environmental Cleanliness

Artificial Intelligence Course Project | Self-Collected Image Dataset | 2026

Field	Details
Project Title	AI-Based Garbage Detection System Using CNN
Course	Artificial Intelligence
Submission	2026
Member 1	Adan Zubair Roll No: BSCS24011
Member 2	Fizza Tahir Roll No: BSAI24018
Dataset	300 Self-Collected Images (Garbage / Clean Areas)
Model	Convolutional Neural Network (CNN)

1.1 Motivation

Waste management has become one of the most critical environmental challenges of the 21st century. Rapid urbanization and population growth have dramatically increased the volume of garbage generated daily in cities, towns, and residential areas. In developing countries like Pakistan, garbage is frequently found on streets, near parks, and in public areas due to a lack of automated monitoring and awareness. Traditional waste management systems rely entirely on manual inspection and labor, making them slow, inconsistent, and inefficient. Artificial Intelligence — specifically deep learning using Convolutional Neural Networks — offers a powerful and practical solution to automate garbage detection directly from camera images.

Key Challenges:

Challenge	Description
No Automation	Traditional systems require manual monitoring with no real-time detection.
Visual Complexity	Garbage varies in size, color, shape, and type, making detection hard.
Lighting Variation	Outdoor images have different lighting — day, night, cloudy conditions.
Background Clutter	Garbage is often mixed with roads, grass, or other non-waste objects.
Limited Local Datasets	Few publicly available datasets represent garbage in local environments.

Applications of AI-Based Garbage Detection:

- **Environmental Monitoring:** Automatically identify polluted areas using camera feeds or drone images.
- **Smart City Management:** Help city authorities prioritize garbage collection routes in real time.
- **Public Awareness:** Display detected garbage on public dashboards to encourage cleanliness.
- **Waste Segregation:** Classify garbage type (plastic, organic, paper) for better recycling.
- **UAV Surveillance:** Deploy on drones to monitor large, remote, or hard-to-reach areas.

1.2 Problem Statement

Problem Statement

"Develop an AI-based system using Convolutional Neural Networks (CNN) that automatically detects garbage in self-collected images to support environmental cleanliness and public awareness."

2 Background / Literature Review

The following four recent research works cover AI-based garbage detection and classification using deep learning techniques.

Paper 1 (2026): Garbage Detection Algorithm Based on Deep Learning (Mask R-CNN)

Global Journal of Engineering Innovations & Interdisciplinary Research

- Proposed Mask R-CNN for detecting and segmenting multiple types of garbage in a single image.
- Model was tested in complex outdoor environments with variable lighting and cluttered backgrounds.
- Achieved high segmentation accuracy and proved deep learning is effective for real-world waste detection.
- Gap: Model is computationally heavy; not suitable for deployment on low-cost devices or mobile phones.

Paper 2 (2024): WasteInNet: Deep Learning for Real-Time Waste Identification (YOLOv7)

ScienceDirect — Computers and Electronics in Agriculture, 2024

- Used YOLOv7 object detection model for real-time identification of various types of waste materials.
- Focused on classification of waste into categories relevant to sustainable waste management.
- Demonstrated that YOLO-based models provide fast and accurate results in real-time applications.
- Gap: Required large labeled datasets that are difficult to collect in local or resource-limited environments.

Paper 3 (2023): Automatic Garbage Detection Using an Optimized YOLO Model

Signal, Image and Video Processing — Springer, 2023

- Proposed an improved and optimized YOLO architecture for garbage detection with reduced computational cost.
- Achieved good detection accuracy with faster inference, making it suitable for real-time embedded systems.
- Tested on street and public area images; model generalized well across different garbage types.
- Gap: Optimization trade-offs led to reduced detection accuracy for small or partially occluded objects.

- Combined CNN-based detection with drone surveillance to identify garbage in large outdoor areas.
- Achieved approximately 94% detection accuracy across diverse environments and angles.
- System enabled monitoring of hard-to-reach areas such as rooftops, forests, and riverbanks.
- Gap: Drone-based systems are expensive and require specialized hardware, limiting accessibility.

Summary: The reviewed literature confirms that CNN, YOLO, and Mask R-CNN are the dominant models for garbage detection. However, most systems are computationally expensive, require large datasets, and do not address local environments. Our project improves by using a simple CNN with a small self-collected dataset from local streets and areas, making it practical, beginner-friendly, and locally relevant.

3 Dataset Details

As per course requirements, the dataset is entirely self-collected using a smartphone camera. No preprocessed or publicly available datasets are used.

3.1 Dataset Overview

Parameter	Details
Total Images	300 Images (Self-Collected)
Class 1 — Garbage	150 images of areas with visible garbage
Class 2 — Clean	150 images of clean areas with no garbage
Image Format	JPEG / PNG
Captured By	Adan Zubair (BSCS24011) & BSAI24018
Collection Method	Smartphone Camera (1080p)

3.2 Collection Locations & Strategy

- **Streets & Roads:** 75 images (garbage + clean) — most common real-world setting.
- **Hostel / Campus Area:** 60 images — captures everyday student-environment garbage.
- **Parks & Open Areas:** 60 images — captures scattered outdoor waste.
- **Near Garbage Bins:** 60 images — includes both overflowing and empty bins.
- **Markets / Bazaars:** 45 images — high garbage density, realistic conditions.

3.3 Data Distribution Breakdown

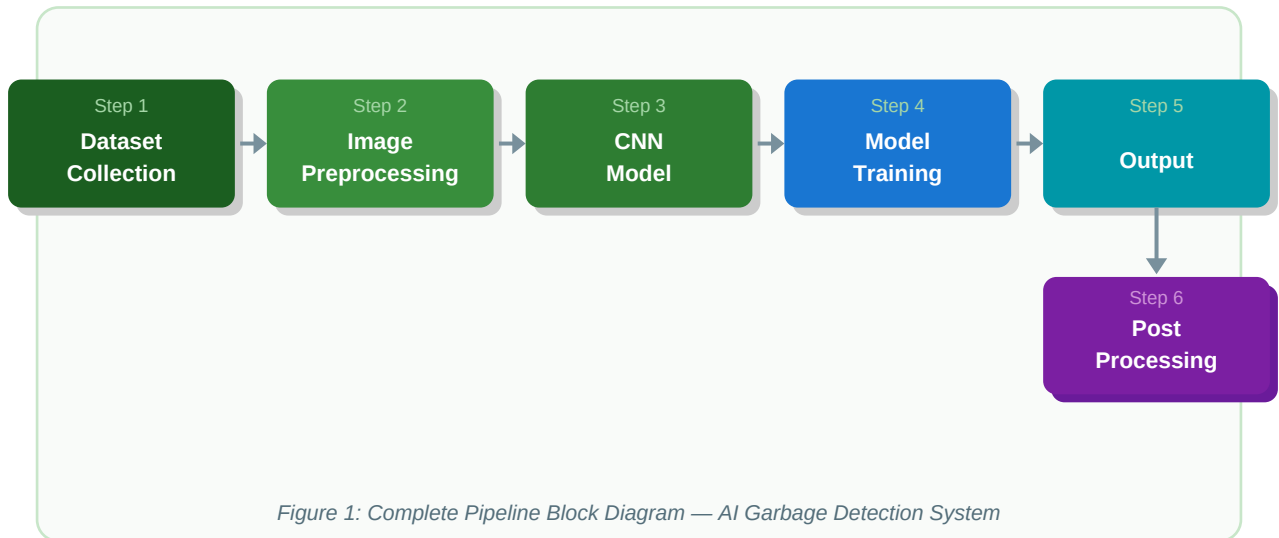
Location	Garbage Images	Clean Images	Total
Streets & Roads	40	35	75
Hostel / Campus	30	30	60
Parks & Open Areas	30	30	60
Near Garbage Bins	30	30	60
Markets / Bazaars	20	25	45
TOTAL	150	150	300

3.4 Preprocessing Steps

#	Step	Tool	Detail
1	Resize	OpenCV / PIL	All images resized to 224 x 224 pixels
2	Normalize	NumPy	Pixel values scaled from [0, 255] to [0.0, 1.0]
3	Grayscale (Opt.)	OpenCV	Optional: convert to grayscale to reduce complexity
4	Data Augmentation	Keras ImageDataGen	Flip, Rotate (30 deg), Brightness, Zoom (0.2)
5	Train/Test Split	scikit-learn	80% Training 20% Testing (240 train / 60 test)
6	Label Encoding	Keras / NumPy	Binary Labels: 0 = Clean, 1 = Garbage

4 Proposed Methodology

4.1 Complete Pipeline Block Diagram



4.2 CNN Model Architecture

A simple custom Convolutional Neural Network (CNN) is proposed, built with TensorFlow/Keras. The architecture is intentionally lightweight and beginner-friendly while being well-suited for binary image classification (Garbage vs. Clean).

Layer #	Layer Type	Configuration	Output Shape
1	Input Layer	224 x 224 x 3 RGB Image	(224, 224, 3)
2	Conv2D	32 filters, 3x3, ReLU	(222, 222, 32)
3	MaxPooling2D	Pool size 2x2	(111, 111, 32)
4	Conv2D	64 filters, 3x3, ReLU	(109, 109, 64)
5	MaxPooling2D	Pool size 2x2	(54, 54, 64)
6	Conv2D	128 filters, 3x3, ReLU	(52, 52, 128)
7	MaxPooling2D	Pool size 2x2	(26, 26, 128)
8	Flatten	—	(86528,)
9	Dense	128 units, ReLU activation	(128,)
10	Dropout	Rate = 0.5 (prevent overfit)	(128,)
11	Output Dense	1 unit, Sigmoid activation	(1,) — Binary

4.3 Training Configuration

Parameter	Value / Detail
Optimizer	Adam (learning rate = 0.001)
Loss Function	Binary Cross-Entropy (binary classification)
Epochs	25–30 (with Early Stopping, patience = 5)
Batch Size	16 or 32
Train / Test	240 Training Images 60 Test Images
Metrics	Accuracy, Precision, Recall, F1-Score
Framework	TensorFlow 2.x / Keras (Python 3.x)
Platform	Google Colab (Free GPU) or Local Machine

4.4 Output & Post-Processing

- **Primary Output:** Binary label prediction — "Garbage Detected" or "Clean Area" — with a sigmoid confidence score (e.g., 0.91 = 91% garbage).
- **Accuracy / Loss Plot:** Plot training vs. validation accuracy and loss over all epochs using Matplotlib to evaluate model convergence.
- **Confusion Matrix:** Visualize true positives, true negatives, false positives, and false negatives using a Seaborn heatmap.
- **Classification Report:** Print Precision, Recall, and F1-Score for each class using sklearn.metrics.classification_report.
- **Confidence Score:** Display the model prediction probability alongside the output label to indicate detection certainty.

5

Team Roles & Work Division

Member	Roll No	Role	Responsibilities
Adan Zubair	BSCS24011	Data Collection & Model Development	Collect 150 garbage images, run preprocessing scripts, build and train the model.
Fizza Tahir	BSAI24018	Data Collection & Evaluation	Collect 150 clean area images, apply augmentation, generate confusion matrix and metrics.

Required Tools & Libraries

- **Python 3.x:** Main programming language

- **TensorFlow / Keras:** CNN model building, training, and evaluation
- **OpenCV / PIL:** Image reading, resizing, and preprocessing
- **NumPy:** Array operations and pixel normalization
- **Matplotlib / Seaborn:** Plotting accuracy curves and confusion matrix heatmap
- **scikit-learn:** Train-test split and classification metrics
- **Google Colab:** Free cloud-based GPU for faster model training

This project is practical, achievable, and well-scoped for a 2-member beginner team. The 300 self-collected images are sufficient for training a binary CNN classifier. The system can be deployed to support local environmental cleanliness and public awareness.